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Unit 4 - Variables

PDF Documents

PDF Document 1: 1738066241502-Chapter 3_unit4 Variables.pdf

This PDF document is included in the unit materials.

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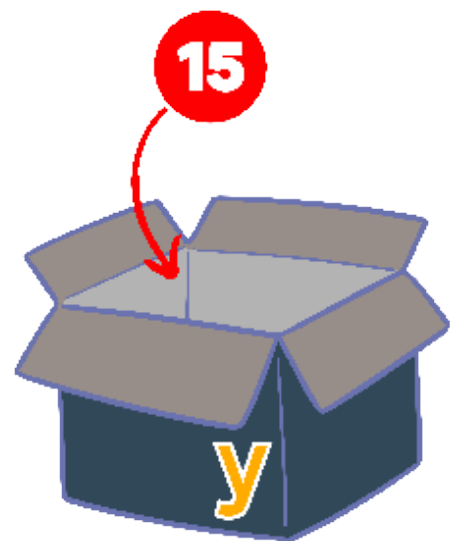
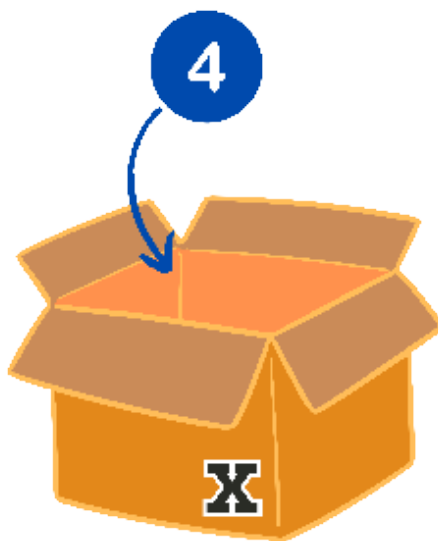


UNIT 4 VARIABLES

Variables

In mathematics, variables such as **x** and **y** are used to represent unknown values that can change. For instance, in the equation $x + 2 = 5$, **x** is a placeholder for a value we need to discover. Once we solve the equation, we find that **x** equals 3.

✓ VARIABLES



Similarly, in programming, a variable is a name assigned to a storage area in the computer's memory where data can be stored, modified, and retrieved later. Just like in math, the value of a programming variable can change as the program runs. For example, if we set a variable **x** in a program with $x = 5$, we can later change it to $x = 10$ based on some conditions or calculations performed during the execution of the program.

EXPLANATION WITH CODE EXAMPLE

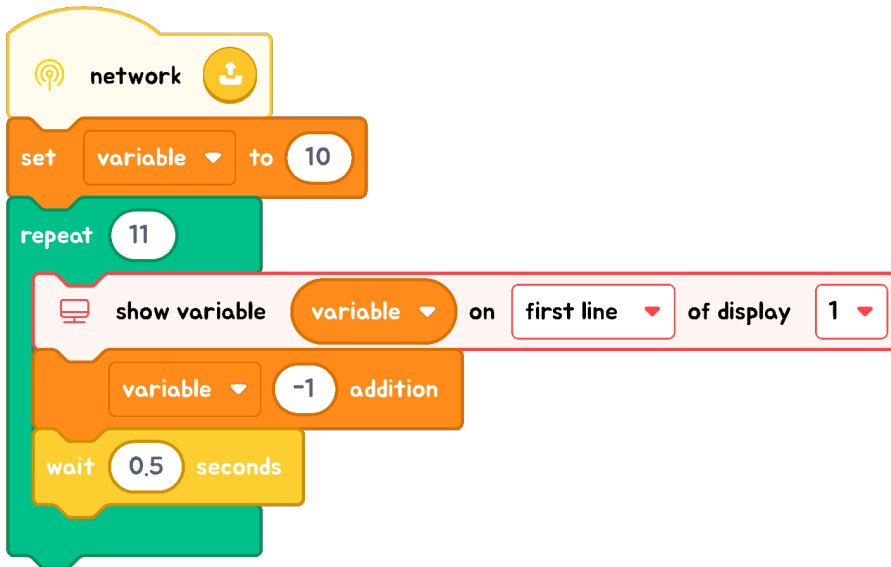
INCREMENT AND DECREMENT OPERATORS

Increment and decrement operators are used to adjust a variable's value by a fixed amount, typically 1, which simplifies coding in repetitive tasks. These operators are especially useful in loops for iterating over sequences.



- The block on the left, known as the increment operator, increases the variable's value by 1.
- The block on the right, known as the decrement operator, decreases the variable's value by 1.

EXAMPLE OF A COUNTDOWN CODE:



Code Insight

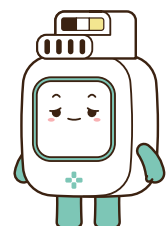
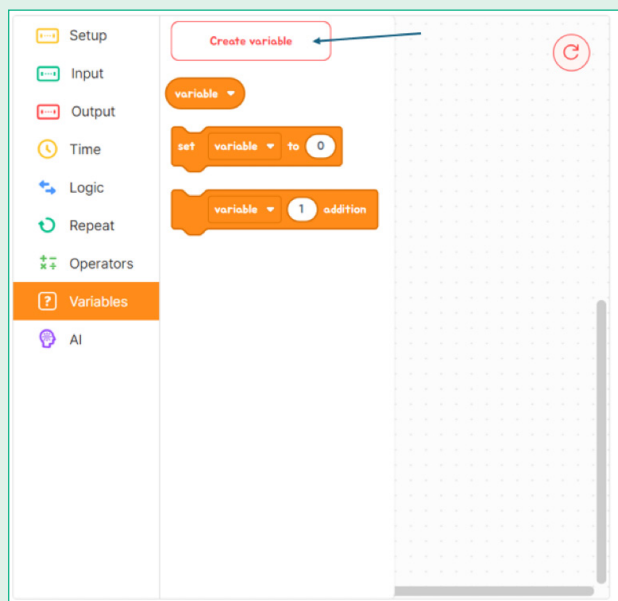
Below is an example of code that counts down from 10 to 0. Here, the decrement operator is used to reduce the value of a variable with each iteration, and this value is displayed on the screen (the Display module).



LET'S MAKE A VARIABLE IN CODE EDITOR:

Step1

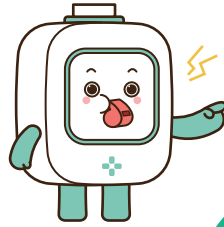
Go to the Variables section and click "Create Variable."





Step2

Type the name for your variable (like "mode").

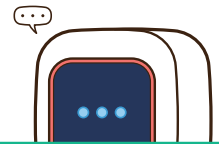


Step3

After typing the name, click "Make" to create it.

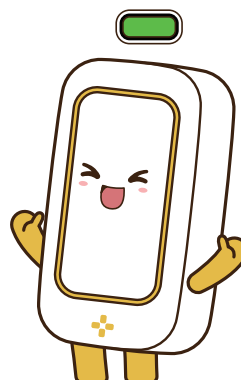
Step4

You'll then see your new variable in the list.



Step5

Click on the area where variables are shown to see the one you made.

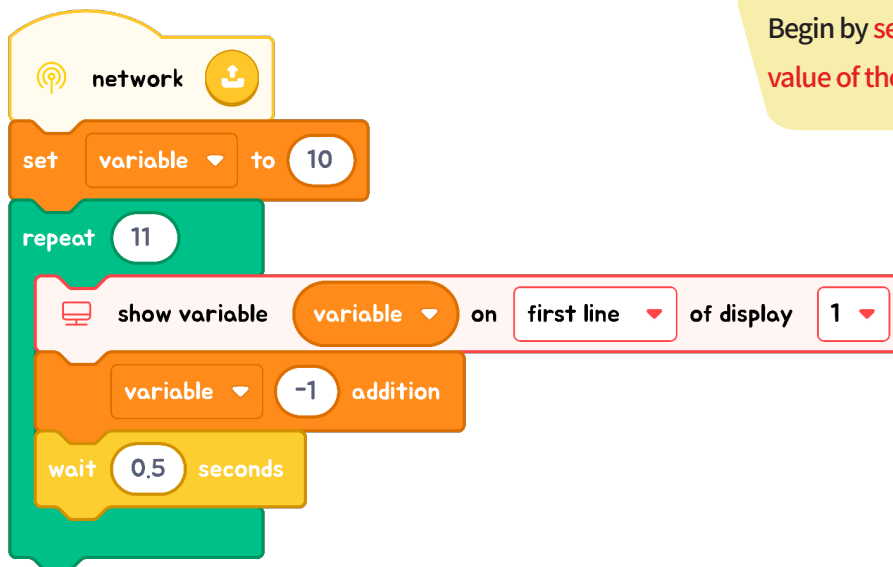


Exercise

Exercise: Modifying the Code to Count Up

In the previous decrement example, the code counts down from 10 to 0. Modify this code to count up from 1 to 10 instead. Remember to set the starting value of the variable appropriately and adjust the loop to increment the variable until it reaches 10.

EXAMPLE OF A COUNTDOWN CODE:



TIPS

Begin by **setting the initial value of the variable to 1.**



Multiple-Choice Question

What is the purpose of a variable in programming?

- ☐ A To store and manipulate data
- ☐ B To perform calculations and store the results
- ☐ C To provide a way to output data to the user
- ☐ D To control the flow of execution in programs